# Homy: Exploring the Design of a Wayfinding Device That Supports Environmental Connection for People With Dementia

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## Abstract

People with dementia often face challenges in navigating outdoor spaces and therefore do not always go outside anymore. This study explores Homy, a voice-based wayfinding device designed to support both wayfinding and meaningful engagement with outdoor surroundings. A field study with four participants and two caregivers showed the potential of Homy to effectively guide users through voice prompts and meanwhile redirected attention to nature and nearby objects, triggering reflection and emotional responses. This paper provides lessons learned and best practices for designing wayfinding devices for people with dementia.

## CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; **Accessibility**.

## Keywords

Cognitive disability, Dementia, Field study, Person–environment interaction, Wayfinding

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## 1 Introduction

Dementia is a rapidly growing global health challenge. In 2021, an estimated 57 million people were living with dementia worldwide, a number expected to triple by 2050 due to population ageing [43]. Dementia is an umbrella term for diseases that progressively affect the brain, often leading to disorientation, memory loss, and reduced decision-making abilities [1, 23]. As symptoms worsen, people with dementia become increasingly dependent on others for daily tasks and care support. Care systems worldwide are struggling to meet this growing demand [36, 42], with staff shortages and financial pressure limiting the time available to provide personal attention and meaningful activities [26]. This often means that staff lack the capacity to accompany residents outdoors or to organise outdoor activities. As a result, many people with dementia living in care homes rarely go outside [3, 24, 32].

This is problematic, as being able to go outside has numerous health benefits [2, 4, 8, 40, 41].The combination of these well-documented benefits and the limited opportunities for outdoor activity in dementia care highlights the need for assistive technologies that enable people with dementia to go outside safely and independently. Much design research has focused on GPS-based tracking systems that can reassure caregivers [14, 18, 28, 35, 39, 44]. A more recent line of research investigates tools that support users' wayfinding abilities [5, 9, 19, 21, 31]. Hereby, wayfinding can be understood as "the cognitive process of finding one's way based on an understanding of the environment" [34, p. 247]. While wayfinding support alone can already be highly valuable for people with dementia, many existing systems approach wayfinding as a purely functional navigation task [12]. This task-oriented framing risks overlooking the experiential qualities of walking, such as sensory engagement, emotional comfort, and a sense of presence in the environment [22].

In this extended abstract, I challenge the assumption that wayfinding devices for people with dementia exist solely for navigation. Instead, I propose a perspective that treats wayfinding as an experience rather than a task. Walking can be an opportunity to connect with one's surroundings and enjoy the present moment. Within this view, wayfinding remains a key system function, but it is complemented by features that encourage awareness of environmental elements such as weather, trees, buildings, and other people. This work investigates the research question: How can a wayfinding device support people with dementia in navigating while also encouraging presence and meaningful engagement with their surroundings? This question has two components: how a device can enhance awareness of the environment and how doing so might influence the functional quality of wayfinding support.

## 2 Related Work

Spending time outdoors has well-documented benefits for people with dementia. Research shows that walking in natural environments can reduce stress [40] and loneliness [13], while supporting mobility [45] and improving sleep [8]. Despite these benefits, walking independently can also be a challenge. Around 70 percent of people with dementia go missing at least once [37], largely due to impairments in spatial navigation and remembering environmental cues [16, 33, 38]. As a result, people with dementia are discouraged from going out alone, limiting opportunities for meaningful engagement with the outdoor environment [30].

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To address safety concerns, many technological solutions rely on GPS-based tracking devices [6, 10, 12, 35, 39]. These systems allow caregivers or family members to monitor the user's location, with the goal of reducing worry and improving safety [11]. While these devices can increase this sense of security for caregivers, they raise several ethical and experiential concerns. Questions emerge about informed consent, autonomy, and the fairness of continuous monitoring [20]. Beyond ethics, many GPS-based systems prioritize the needs of caregivers rather than the experiences of the person with dementia. They are effective for locating someone who is lost but do little to prevent disorientation in the first place, nor do they offer reassurance or a sense of agency to users [12].

A second line of research focuses on supporting the wayfinding abilities of people with dementia [5, 9, 17, 19, 21, 31]. These technologies aim to enhance autonomy by providing guidance that builds on the user's own cognitive abilities. Unlike GPS trackers, wayfinding technologies promote user participation by offering directions or environmental cues designed to support navigation.

The technologies described above represent significant progress in addressing safety and orientation needs for people with dementia. However, these systems tend to prioritize functional goals such as reaching a destination, while paying less attention to experiential needs. For many people with dementia, going outside is not only about arriving somewhere, but also about enjoying fresh air, sensory stimulation, and observing their surroundings [22, 29]. While accurate guidance remains essential, this suggests a need for technologies that also support presence, curiosity, and meaningful interaction with the environment.

## 3 Prototype Design

**Figure 1: Homy up close**

**Figure 2: Homy in use**

Homy was developed through a co-design process involving people with dementia and professional caregivers. It provides gentle guidance during walks while encouraging engagement with the surrounding environment. To achieve this, Homy is a wearable wayfinding device that delivers voice cues about the environment during outdoor navigation. The device is 3D-printed in the shape of a flower and contains a Bluetooth module, speaker, battery, and an on/off button.

A supporting web-based application was designed for caregiver use to initiate the walks. The interaction flow of the application was intentionally kept short and simple to facilitate seamless integration

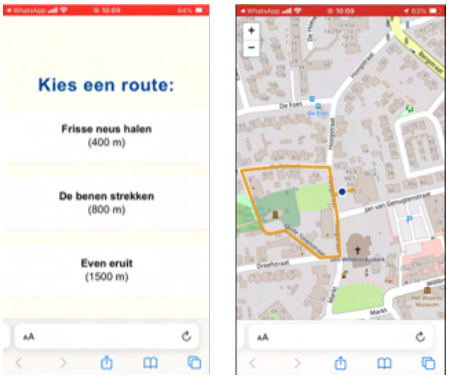


**Figure 3: The web application**

into the caregiver's existing workflow. Upon opening the app, the caregiver selects a preferred walking distance: short, medium, or long. After this, the phone is connected to Homy via Bluetooth, and the user can begin the walk. During walking, the user received short spoken direction prompts, such as "turn left here". Insights from people with dementia involved in the co-design process indicated that these prompts needed to be simple, concise, and delivered at the right moment to minimize cognitive demands. In addition to direction promps, Homy also provides environmental prompts using three types of data:

- Municipal spatial planning data from the Dutch BGT [15], which provided information on objects like trees, benches, and sports facilities. An example prompt is: *"What a beautiful tree over there. It looks very old."*
- OpenStreetMap data [27] for context about buildings, such as churches or restaurants. For example: *"Ah, there's a restaurant. It's so nice to have places to eat nearby."*
- Real-time weather data to comment on current weather conditions. For example: *"What a lovely sun is shining today, isn't it?"*

Using these combined data, Homy anticipates what the user is likely to see and respond with predefined prompts to redirect attention to the environment. To regulate the timing of these messages, a simple rule-based algorithm was developed. It ensures that prompts are neither delivered too frequently nor separated by long periods of silence. The system also refrained from saying environmental prompts shortly before or after directional instructions to prevent distraction at critical moments.

## 4 Methods

The study was approved by the ethics review board of Eindhoven University of Technology.

Four participants with dementia were recruited in collaboration with a care home, with two caregivers assisting in participant selection and later interviewed for additional insights about the participants.

Walks were taken in the neighborhood around the care home. To ensure participants' safety, the researcher and a formal caregiver were always nearby to provide support if needed. Caregiver interviews were held indoors in the care home.

**Table 1: Participant overview**

| Participant | M/F | Stage of dementia | Walking roller? |
|---|---|---|---|
| Participant 1 (P1) | F | Mid | No |
| Participant 2 (P2) | F | Mid | No |
| Participant 3 (P3) | F | Mid | No |
| Participant 4 (P4) | M | Early | Yes |

The field study consisted of three parts: 1) an introduction in which participants received study information, provided informed consent, and were shown Homy's functions; 2) a walk-along interview [25][34] where each participant completed a 450-meter route using Homy and was encouraged to think aloud to get insights into their observations, feelings, and decisions; and 3) a short closing interview in which participants reflected on their overall experience. By conducting the field study in participants' everyday outdoor environment, I aimed to increase the ecological validity of the findings and capture wayfinding behavior under near-natural conditions.

The two caregivers participated in semi-structured interviews about their experiences with the residents, wayfinding challenges, and technology use.

All data was recorded using a phone and transcribed by the researcher. A thematic analysis was conducted to gain insight into the qualitative data. Using NVivo, the data was clustered into themes and subthemes through an inductive process, following the approach proposed by Virginia Braun and Victoria Clarke [7]. This analysis identified the themes and subthemes presented in Figure 4.

**Figure 4: The themes and subthemes of the thematic analysis**

# 5 Findings

## 5.1 Wayfinding Performance

*5.1.1 Getting successfully from A to B.* It was observed that all participants successfully returned to the starting point using Homy, without external help. The wayfinding prompts were followed accurately, with no observed confusion. P3 appeared to express a preference for concise prompts delivered at the right moment, rather than receiving detailed instructions in advance: *"Yes, in small quantities. Because you don't have to tell me that I have to go that way in a moment.... Then it also becomes too complicated, too much."* The concise directional prompts that were used by Homy were furthermore described as *"clear"* by P1, P2 and P4.

*5.1.2 A sense of reassurance and comfort while moving from A to B.* While Homy effectively brought all participants back to the starting point, an important question remained whether participants themselves felt confident and reassured. All participants appeared relaxed and comfortable. P1 shared, thereby, the following: *"But luckily, there is then something that shows me the way back. That is a bit of reassurance."*

Furthermore, Homy might have the ability to help reduce stress by stimulation and steering the focus away from wayfinding. According to C1, uncertainty and silent moments can be deeply unsettling for some residents: *"The empty time. One resident literally says: I panic because then there is a black hole. I don't know what is coming."* Hereby, C2 shared a tactic that works well for getting residents at ease: *"I often change the subject. Then suddenly that emotion is gone."* Homy's environmental prompts seemed to serve this function, keeping the participants engaged and less focused on potential anxiety.

## 5.2 Supporting Environmental Connection using a Wayfinding Device

*5.2.1 Voice-Based Interaction: A low-threshold opportunity for engagement with the environment.* Homy's voice-based approach seemed to fit well with the abilities of the participants. It did not require many digital or fine motor skills, it just needed someone to listen. P1 shared: *"OK, so here in the building they already switch it on and then it just guides me completely by itself? Then I don't have to do anything anymore, except listen? Oh, then I would really like that. Because I am not a hero with such things."*

As for how they interacted with Homy, all participants experienced it as nice to have someone walking with them. P4 for example described it as "convivial". For three out of four participants, the interaction with Homy seemed to go beyond passive listening. It was observed that they started to talk back, as if walking with a companion. When Homy said goodbye at the end of a walk, P1 smiled and replied that she had also enjoyed walking together. When the device made a comment about the rain, P4 paused and responded: *"It is indeed not so nice weather isn't it?"*

*5.2.2 Participant's reactions to environmental prompts.* Observations across all participants indicated that prompts about the environment had a clear ability to direct the focus of the user. When Homy pointed out something nearby, all participants would often look at it. Two types of prompts emerged as especially effective. The first type included prompts about aesthetically pleasing objects

from nature or architecture. These prompts didn't just make participants look; they seemed to encourage them to think about their beauty. When Homy pointed out a nearby church, P2 responded: *"Isn't that beautiful? So pretty between those trees!"*

The second type of prompt that seemed to work well were those that triggered a memory or story. These prompts often led to deeply personal responses. When Homy mentioned a sports facility next to the path, P3 recalled playing pétanque earlier in life and began telling a story about it. Another example involved Homy commenting on the charming look of cobblestones in the street. P1 later reflected: *"That made me think of how it used to be normal in the past. It gave me a happy feeling. Those cobblestones aren't around much anymore, but I got to see them again."* This quote illustrates that participants did not just notice the prompts, they sometimes even responded emotionally. In fact, for all participants, smiling and positive emotions were observed at some point during the environmental prompts.

However, observations also showed that prompts that referred to less meaningful or visually striking objects, such as a bicycle rack or a hedge, generally did not receive a clear response from the person with dementia.

## 6 Discussion

*Opportunities of voice-based interactions to enrich wayfinding experiences:* Homy's voice-based interaction appeared to be one of its strongest features. Spoken prompts offered a low-threshold, intuitive way of engagement that aligned well with the abilities of the participants with dementia. The presence of a voice also created a subtle sense of companionship, with several participants replying to the device as if it were a walking partner.

*The reactions on environmental prompts:* The findings showed how environmental prompts directed participants' attention toward their outdoor surroundings. Building on prior work that highlights the cognitive and emotional benefits of outdoor activity for people with dementia [8, 13, 40, 45], increased attentiveness to the surrounding environment may help reinforce these positive effects. While this study did not measure these effects directly, it indicates that brief audio prompts may help create the conditions under which such benefits occur.

*Wayfinding performance:* Participants' successful navigation from A to B suggests that voice-based guidance can be sufficient for wayfinding, supporting some prior work on assistive navigation [9, 21, 31]. Importantly, this study extends earlier task-oriented approaches by also highlighting the role of reassurance during navigation. By reducing silence and uncertainty through gentle prompts, Homy may help alleviate discomfort, a too often-overlooked factor in wayfinding [12, 20]. Potentially, this sense of reassurance contributes to wayfinding, but this relationship was not examined in depth and remains an important topic for future research.

*Autonomy vs. safety:* While no system can guarantee full safety for people with dementia walking alone, the findings of this study suggest that Homy can strengthen arguments within the ongoing debate between autonomy and safety. Participants were able to navigate successfully from A to B, but more importantly, the device appeared to foster a stronger connection with the outdoor environment, reinforcing the benefits of outdoor walking [2, 40, 41]. By placing greater emphasis on environmental engagement, Homy demonstrates the value of independent outdoor walking for people with dementia as a meaningful and beneficial practice within dementia care.

## 7 Conclusion

This study explored how a wayfinding system like Homy can support people with dementia not only in navigation but also in engaging meaningfully with outdoor surroundings. The findings indicate that simple voice prompts can successfully assist users with dementia in wayfinding while also redirecting attention outward, evoking memories, and fostering emotional responses. Participants reacted positively to cues about nature and buildings, indicating that environmental prompts can enrich the walking experience.

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ChatGPT was used solely to restructure some individual sentences (not entire texts). For this purpose, the following prompt was used: "Rewrite this sentence to improve the sentence structure."

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